GNN-BERT Music Context Understanding: A Hybrid Graph Neural Network and BERT Framework for Multimodal Music Analysis

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Code: `https://github.com/auritro25/gnn-bert-music-context`

Abstract

Music is a complex concept that covers several aspects beginning from melody and rhythm to timbre and description semantics. The paper talks about GNN-BERT, the hybrid system developed to understand the context of music by applying Graph Neural Network (GNN) encoders, which are used over the generated audio graphs, and BERT-based text encoders, which are used to encode comments and tags. The system is tested on four tasks of different levels of complexity including (1) BERT-based multi-label tagging of captions in MusicCaps; (2) two different techniques for music genre classification (GNN vs CNN) on FMA-small data; (3) GNN-BERT technology to unite the data obtained through audio classification and textual description of music; and (4) application of comparative GNN-BERT analysis on MusicCaps.

The text-only BERT model significantly beats the audio-only and combined models, reaching a score of 0.577 for Task 3, while the latter ones get only between 0.031 and 0.188. The results imply that the semantic captions are much more valuable than the audio graph representation in this workflow. At the same time, the fusion model enhances performance compared to the unimodal systems on FMA-medium. The contrastive retrieval system is also able to retrieve relevant matches in the top 10 results around 20% of the time in either direction. We present full details of the evaluation in the section below and analyze discrepancies between audio and text results along with future improvements in the audio graphs, staged fine-tuning, and contrastive pre-training.

1 Introduction

Computational analysis of music involves processing numerous interdependent types of information: the acoustic characteristics of the music (melody, harmony, rhythm, timbre), and the meaning attached to those characteristics by people (genre, mood, other textual descriptions). Convolutional and recurrent networks used with audio spectrograms work well, as they can identify local linear patterns of sound; however, they do not identify relationships between subsections of a musical work (e.g., a chorus and a stanza, or two noncontiguous segments with similar timbres). For this reason, it makes sense to use Graph Neural Networks (GNNs) in this situation, since a music composition can be represented as a graph of segments (or chords), with edges representing their adjacency and similarity.

At the same time, language models based on the transformer architecture (BERT, etc.) have been widely adopted as one of the best methods for encoding free text into vectors. The combination of a graph encoder representing audio information and a language model for textual information can

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provide a solution for music analysis by using both types of data rather than separating them into two distinct problems.

The GNN-BERT Music Context Understanding project operationalizes an approach across four difficulty levels of its tasks: (1) a baseline BERT model, based on text data, for multi-labeling task from string descriptions; (2) a comparison of CNN and GNN in audio-based genre identification; (3) a fusion approach, where GNN-processed data is combined with BERT data to provide genre and tag predictions; and (4) a contrastive dual encoder methodology, which achieves better alignment of audio graphs and string literals. We present outcomes obtained for the four tasks according to the leakage-controlled evaluation principles and describe the strengths and weaknesses of the proposed approach.

Contributions. Contributions made by the researchers are as follows: (a) a full pipeline of sound data processing and data evaluation, based on audio and textual data popular algorithms; (b) comparison of four distinct fusion strategies on the basis of the same data; (c) design of a contrastive retrieval model with Recall@K evaluations in both directions; and (d) an honest assessment of the differences between audio and text modalities along with the specific suggestions regarding future studies in this field.

2 Related work

The study is based on three previous works.

Text encoders for music. Pre-trained transformer language models, like BERT, are now frequently utilized for encoding musical event descriptions, tags, and lyrics through free-text channels into fixed-size contextual embeddings compatible with downstream classifiers. The BERT model is pre-trained using a classification head masking the [CLS] token.

Graph neural networks for representing sounds. The idea behind employing GNNs, and especially GraphSAGE-style neighborhood aggregation techniques, is to model the structure of relations existing in sequential and structured data by using (temporal or similarity) relations between nodes; we borrow this idea of representing track in the graph of sound events.

Multimodal fusion and contrastive alignment. To implement an integration process, different types of encoders are used in accordance with either concatenation or attention methods to create models that operate with information from both modalities or allow retrieving from corresponding modalities (e.g., InfoNCE models). In the course of the implementation of Tasks 3 and 4, our fusion methods and models were designed according to the requirements of the described methodology.

3 Datasets

The authors utilized a total number of three data sets which are publicly available and each of them represents a different modality and task.

The first one is named MusicCaps. It is a set of music clips with relevant information described by experts in free text. This collection is the only one being involved in the first and fourth tasks listed in the research, namely caption-based multi-label tagging and contrastive music-text retrieval.

The second dataset is called FMA-small which is a part of the Free Music Archive and contains labeled music samples. It is used in the second task of the study that comprises two models: CNN that works with the audio samples and GNN which relies on constructed audio segment graph.

FMA-medium is a bigger part of the Free Music Archive used in the third task of multimodal experiment in which the authors compared graphs and text representations used for tagging purposes.

4 Method

4.1 Pipeline overview

The system pipeline follows five standard stages that are consistently used in all four tasks: (1) audio preprocessing, (2) graph construction, (3) GNN encoding, (4) text encoding, (5) task-related fusion, classification, or contrastive alignment.

Audio preprocessing. The raw audio is resampled and converted to mel-spectrogram, MFCC, and chroma features. The tracks are broken down into fixed or beat-synchronized segments and the features are normalized for each track.

Graph construction. Each track is transformed into a graph where the audio periods (or chords) are the nodes with the audio features extracted from audio being the features of the respective nodes. The edges represent time-ordered connections between neighboring segments as well as similarity connections to segments that share some feature vector similarities that exceed a selected threshold. This representation of the graph is utilized to reflect the structure of relations between segments (e.g. the segments that are repeated or harmoniously linked) which cannot be captured by the standard CNN/RNN model working with the spectrograms alone.

4.2 GNN encoder

A GraphSAGE-style neighborhood aggregation process is employed in a graph encoder. For layer l, the representation of each node i is updated through aggregation of the representation of its neighbors $N(i)$:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \cdot \mathrm{AGG}\left(h_i^{(l)}, \{h_j^{(l)} : j \in N(i)\}\right)\right) \tag{1}$$

where $h_i^{(0)}$ is derived from the audio features at the segment level and AGG is a mean aggregation of representation of the neighbors. The graph-level embedding is computed through mean pooling of all the representations after L layers of message passing.

$$g = \frac{1}{|V|} \sum_{i \in V} h_i^{(L)}.$$

4.3 Text encoder

The captions and tags are tokenized and sent through the pretrained BERT encoder. Here, the encoding of the [CLS] token is used as a fixed-length text embedding, $t = \mathrm{BERT}_{\mathrm{CLS}}(X_{\mathrm{text}})$. For Task 1 baseline, this value t is given to a linear classification layer with a sigmoid activation function; for the Task 3 and 4, t (or all token embeddings together H_{text}) will be fused with the graph embedding g.

4.4 Fusion strategies (Task 3)

Four versions were trained and tested on the same FMA-medium fusion task: (i) BERT-only, which used only the text embedding t; (ii) GNN-only, which took input as only the graph embedding g; (iii) early concatenation, $z = \mathrm{CONCAT}(g, t)$, which uses a fusion process with a linear classification head; and (iv) learned fusion, which combines g with H_{text} through fusion layer and then through the classification head. In all the variants, prediction is expressed as $\hat{y} = \sigma(Wz + b)$, and training is performed by minimizing multi-label binary cross-entropy.

$$\mathcal{L} = -\frac{1}{K} \sum_{k=1}^{K} \left[y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)\right] \tag{2}$$

4.5 Contrastive retrieval (Task 4)

By making use of the dual-encoder-based model in cross-modal retrieval, it is feasible to train the model by means of using a normalized graph embedding, g_i, and normalized caption embedding, t_i, for every clip i. The InfoNCE loss facilitates the process of matching pairs throughout the process of training.

$$\mathcal{L}_{\mathrm{NCE}} = -\log \frac{\exp(\mathrm{sim}(g_i, t_i)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(g_i, t_j)/\tau)} \tag{3}$$

The function $\mathrm{sim}(u, v) = u^\top v/(\|u\|\|v\|)$ is introduced here, with τ being a temperature hyperparameter. For evaluation purposes, retrieval is measured using Recall@K, considering both directions of retrieval: text-to-music and music-to-text. Evaluation of the same contrastive encoders had taken place on the zero-shot tag prediction task, where tag relevance is determined via retrieval similarity between the music embedding and embeddings of potential candidate tag phrases, without any specific training related to this task.

4.6 Baselines

There are two audio-only baselines provided for comparative purposes. In Task 2, CNNs applied directly to mel-spectrogram features are used as the baseline for GNNs. In Task 3, BERT-only and GNN-only variants serve as unimodal baselines against which the fused models are compared.

5 Experimental setup

All four tasks were conducted and assessed in the same repository (`https://github.com/auritro25/gnn-bert-music-context`) where leakage control methodology was implemented – and the data that is divided between the training and testing sets was designed in a way that there is no leakage of either artist or track from one set to the other. Thus, the comparison program `scripts/compare_strict_compliance.py` calculates various metrics such as Macro-F1, Micro-F1, and Macro/Micro AUC-PR for Tasks 1 and 2 as well as accuracy scores, while Task 3 receives the same metrics for all four variations of fusion algorithms. Finally, Task 4 provides Recall@1/5/10 for both ways of retrieval in addition to zero-shot AUC-PR as well as F1@top-K metrics for zero-shot tag prediction.

6 Results

6.1 Task 1: BERT multi-label caption tagging

The outcomes of the BERT tagger on MusicCaps, as per the leakage-controlled protocol, are presented in Table 1. The macro-F1 and micro-F1 for the model are recorded as 0.577 and 0.629, respectively, while Macro/Micro AUC-PR is marked as 0.603 and 0.685, respectively. The most favorable result among all four tasks is achieved by the model.

Table 1: Task 1 – BERT multi-label caption tagging on MusicCaps.

Model	Macro-F1	Micro-F1	Macro AUC-PR	Micro AUC-PR	Accuracy
BERT	0.577298	0.629232	0.603319	0.685060	–

6.2 Task 2: genre classification (CNN vs. GNN)

According to Table 2, there is a comparison of CNN's use as the baseline and the encoding thus of GNN. The CNN is proved to be much better in all three metrics of measurement, achieving 0.545 of macro-F1 and 0.541 of accuracy, while GNN scores a mere 0.364 and 0.353 correspondingly. Hence, it can be said that the segment-graph description is still inferior to an ordinary CNN arrangement based on the spectrogram.

6.3 Task 3: GNN-BERT fusion

There are four fusion options shown in Table 3 on the FMA-medium dataset. The proposed fusion scheme exhibits the highest macro-F1 score of 0.188 and micro-F1 score of 0.182 when compared

Table 2: Task 2 – genre classification on FMA-small.

Model	Macro-F1	Accuracy
CNN	0.544607	0.54125
GNN	0.363634	0.35250

to the other alternatives like BERT-only, which obtains a macro-F1 score of 0.179, and early fusion method with a macro-F1 score of 0.178. The ranking in this experiment, where GNN-only scores lowest with 0.031 macro-F1 score, shows that the BERT-only method is superior and proves the outcome of Task 2: that the textual feature is significantly more useful than the audio-feature and its graph alone. It is worth noting that the outcome of Task 3 is lower in comparison to the results of Task 1, due to the higher complexity of the task.

Table 3: Task 3 – GNN-BERT fusion ablation on FMA-medium.

Variant	Macro-F1	Micro-F1	Macro AUC-PR	Micro AUC-PR
BERT-only	0.178989	0.109479	0.211073	0.104344
GNN-only	0.030667	0.027357	0.024215	0.012775
Early concatenation	0.177994	0.122549	0.225460	0.085275
Learned fusion	0.188290	0.181818	0.263020	0.127746

6.4 Task 4: contrastive GNN-BERT retrieval

As shown in Table 4, the contrastive retrieval model is evaluated in terms of Recall@K for the MusicCaps test data in both retrieval directions. The performance of text-to-music retrieval is given as R@1 = 0.0315, R@5 = 0.1243 and R@10 = 0.2102. The results of music-to-text retrieval are somewhat similar to those of the text-to-music retrieval, by which we mean that R@1 = 0.0298, R@5 = 0.1278 and R@10 = 0.1979. The two retrieval directions are rather symmetric in comparison with each other, showing that the model manages to get the right match among the first ten proposals in about one fourth of the attempts performed, which is far better than expected if the ranking were carried out wholly at random, given the size of the test data and the retrieval task.

Table 4: Task 4 – contrastive retrieval Recall@K on MusicCaps (test split).

Direction	R@1	R@5	R@10
Text $\rightarrow$ Music	0.031524	0.124343	0.210158
Music $\rightarrow$ Text	0.029772	0.127846	0.197898

6.5 Zero-shot tag prediction

The same contrastive encoders were tested for zero-shot labeling, without undertaking any task-specific training: Macro AUC-PR = 0.010987, Micro AUC-PR = 0.005899, Macro F1@top-K = 0.008988, and Micro F1@top-K = 0.011119. The results are much lower than those from both the Task 1 supervised BERT tagger and the Task 3 supervised fusion model, which is not surprising since the contrastive encoders were specifically optimized for retrieval alignment rather than for the specific tag vocabulary used in this work. Therefore, the finding should be regarded as a lower bound for the performance of a purpose-built zero-shot classifier exploiting the same embeddings space.

6.6 Summary across tasks

Table 5 collects the headline metric from each task for a single overview.

Table 5: Headline results across all four tasks.

Task	Headline metric	Value
Task 1 (BERT tagging, MusicCaps)	Macro-F1	0.577
Task 2 (CNN, FMA-small)	Accuracy	0.541
Task 2 (GNN, FMA-small)	Accuracy	0.353
Task 3 (Learned fusion, FMA-medium)	Macro-F1	0.188
Task 4 (Text→Music retrieval)	R@10	0.210

7 Discussion

In all three supervised tasks (1-3) some trends can be distinguished. First, text (BERT) part is always stronger than audio-graph (GNN) part: also, it is known that BERT-only in Task 3 performs almost 6 times better than GNN-only (regarding macro-F1) and that in Task 2 CNN method beats GNN by the same margin. At the same time, the combination of methods does not bring outstanding results: learned fusion allows BERT-only method to show 5% better relative macro-F1 and 66% better relative micro-F1 in Task 3, which indicates that although graph representation is weak and does not show any superiority on its own, there is some room for improvement in terms of gaining higher output with its help.

The explanation for the weak performance of the audio-graph module is that graph generation using techniques like fixed window, beat synchronous graph segmentation using temporal adjacency nodes and feature similarity edges over mel/MFCC/chroma features results in a coarse representation as compared to the representations learnt by CNNs directly from the input spectrograms. Also, the GNN encoder has a smaller parameter size and effective receptive field size in raw signals than the CNN's baseline model in Task 2, which may possibly lead to the difference in the performance of the two models in that task. The picture from the retrieval tasks, however, does look different. The Recall@10 score of 0.20 indicates that the contrastive space has genuine, non-trivial cross-modal information.

8 Limitations

When interpreting the outcomes presented here, several constraints should be taken into account. The GNN encoder shows relatively poor standalone performance, which restricts how well fusion models can use the audio-graph branch. The next stages involve working on improving graph construction methods, such as using different kinds of edges, building chord-aware graphs, or developing datasets with beat-synchronized data. The reported results come from a single run of each configuration without averaging them over runs with different seeds, hence, no formal measurements of variance or significance of differences between the fusion methods can be done. FMA-medium and FMA-small datasets do not perfectly represent different genres in a balanced way, and MusicCaps dataset, which is not very large, also limits the generalization of the obtained results. Lastly, the zero-shot tag-prediction test is conducted using embeddings that were not optimized for this particular purpose, therefore, their poor result is a result of discrepancy between training and evaluation goals and not due to the underlying retrieval model.

9 Conclusion

The report described GNN-BERT which is a type of a hybrid system. It is a combined device that integrates BERT-based encoding of texts and graph-based encoding of audio in four tasks. These tasks are: tagging based on the description, genre classification by audio only, cross-modality retrieval, and fusion of modalities. Results indicate that the text modality was more efficient. However, GNN-BERT reached a higher level of performance than previous systems in Task 3. One of the main directions for improving it is to learn how to perform modeling of audio graph better so that the system could work more efficiently.

Code, configuration, and results for all four tasks are available at `https://github.com/auritro25/gnn-bert-music-context`.

References

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NeurIPS Paper Checklist

1. **Claims.** Answer: [Yes]. Justification: The abstract and introduction state that the system is evaluated on four tasks and report the headline macro-F1/Recall@K numbers that are reproduced in Section 6, including the noted weakness of the audio-only branch.
2. **Limitations.** Answer: [Yes]. Justification: Section 8 discusses the GNN encoder's weaker standalone performance, the lack of multi-seed variance estimates, dataset balance/size constraints, and the objective mismatch underlying the zero-shot evaluation.
3. **Theory assumptions and proofs.** Answer: [N/A]. Justification: The paper does not present formal theoretical results requiring proofs; Section 5 gives the model equations used in implementation, not theorems.
4. **Experimental result reproducibility.** Answer: [Yes]. Justification: Section 5 describes the full pipeline (audio preprocessing, graph construction, GNN encoder, text encoder, fusion/contrastive heads) and Section 6 describes the leakage-controlled evaluation protocol and comparison script used to produce all reported numbers.
5. **Open access to data and code.** Answer: [Yes]. Justification: The full implementation, scripts, and results are available at `https://github.com/auritro25/gnn-bert-music-context`. The underlying datasets (MusicCaps, FMA-small, FMA-medium) are publicly available from their original sources.
6. **Experimental setting/details.** Answer: [Yes]. Justification: Section 6 describes the train/test split protocol (leakage control by artist/track) and the metrics computed for each task; per-run hyperparameters are stored alongside the training scripts in the repository.
7. **Experiment statistical significance.** Answer: [No]. Justification: Results are reported from a single run per configuration; no error bars or significance tests are included, as noted in Section 8.
8. **Experiments compute resources.** Answer: [N/A]. Justification: Detailed compute-worker and memory accounting was not tracked separately for this project; experiments were run on a single local development machine.
9. **Code of ethics.** Answer: [Yes]. Justification: The project uses only publicly released research datasets (MusicCaps, FMA) for non-commercial academic evaluation and does not involve human subjects or personally identifying data.
10. **Broader impacts.** Answer: [N/A]. Justification: This is foundational coursework research on music understanding models and is not tied to a specific deployment; no direct path to negative applications beyond those generic to any music tagging/retrieval system was identified.
11. **Safeguards.** Answer: [N/A]. Justification: No pretrained generative model or scraped dataset with a high risk of misuse is released as part of this work.
12. **Licenses for existing assets.** Answer: [Yes]. Justification: MusicCaps and FMA are used under their respective original release terms; both are cited in the References section and in the repository documentation.
13. **New assets.** Answer: [N/A]. Justification: The project does not release a new public dataset; it releases training/evaluation code and result files for the four existing tasks.
14. **Crowdsourcing and research with human subjects.** Answer: [N/A]. Justification: The project does not involve crowdsourcing or human-subjects research.
15. **Institutional review board (IRB) approvals.** Answer: [N/A]. Justification: The project does not involve human subjects research.
16. **Declaration of LLM usage.** Answer: [N/A]. Justification: LLM assistance, where used, was limited to writing/editing/formatting of this report and did not constitute an important, original, or non-standard component of the core GNN/BERT modeling methodology.